

HARD REAL-TIME (MCU)

Deterministic Reflex · 1000 Hz · malloc = 0

Perceive

Fast Sensor Ingestion & Proprioception

IMU · Encoders · PTP Hardware Timestamps

1000 Hz Fast Reflex Loop

Act

1 kHz Safety Reflex & Motor Commutation

Control Barrier Functions · PWM Gate Locks

PRIVILEGE BOUNDARY

SOFT REAL-TIME (SBC / MPU)

Asynchronous Deliberation · 0.5–20 Hz · Linux / NPU

Reason

Semantic Deliberation & Action Chunking

Vision-Language Models · World Models

Untrusted Proposal Generator

Sensory Evidence (Tokens)

Expiring Intent Proposals

